I've spent more time than I'd like to admit staring at a math problem, getting increasingly frustrated, and finally arriving at an answer. Only to find out it was wrong. I've also had days when I felt completely lost but somehow got the next question right. It made me wonder whether how I was feeling could tell me something about how well I was learning.

Reading Hawthorne's research on emotions and cognitive load made me even more curious. If emotions affected how students approached learning, could they also help predict whether a student would answer the next question correctly?

So, with my sister Sanah, I started investigating. I helped formulate our research question, choose Bayesian Knowledge Tracing (BKT) and Deep Knowledge Tracing (DKT) as our approaches, and interpret the results, while she handled the implementation. Using ASSISTmentsdata, we compared predictions based on students' answer histories with predictions that also included estimated affect.

I expected the affect-enhanced models to perform noticeably better, but they didn't. At least, not consistently.

BKT showed small improvements, but they were too small to be practically meaningful. DKT was less predictable, with improvements that varied across experiments.

At first, I wasn't sure what to make of that. I'd spent so much time thinking about how fear, frustration, and confidence affected my own learning that I'd expected those feelings to show up more clearly in the predictions.

But maybe I was asking a narrower question than the one I actually cared about.

We were predicting whether a student would get the *next answer* right. What about whether they would keep trying after getting three questions wrong? Or whether frustration would make them stop asking questions? Those were things I cared about too, but our experiments hadn't tested them.

I also started wondering about the emotions themselves. In our experiments, affect was estimated from learning behaviors. But how much could those estimates really tell us about what a student was feeling?

That's one reason I'm working on Calculus Clinic. It organizes mathematical concepts around their prerequisites, making it possible to examine where a student's understanding might break down. It also provides camera-derived expression scores that could be compared with performance on individual concepts. Those scores are still estimates of emotion, of course. A facial expression doesn't tell us everything going on in someone's head.

Once we have enough data, I want to start by looking at the patterns rather than jumping straight into another model. Do expression scores change when students encounter prerequisite gaps? Are those changes consistent across students? And what, if anything, can we actually conclude from them?

I started this research expecting to find evidence that emotions could make learning predictions better. Instead, I found myself questioning how we measure those emotions in the first place and how emotions affect learning in the short-term and long-term.

I still think emotions matter in learning, but now I have a lot more questions about what our models can tell us about them.

The relationship between positive and painful emotions and cognitive load during an algebra learning task
https://www.sciencedirect.com/science/article/pii/S1041608024001900

Adaptive Knowledge Tracing with Dynamic Memory and Reinforcement Learning:
https://www.mdpi.com/1424-8220/26/6/1878